

Movie Success Factors in the Film Industry (2000-2024): A CRISP-DM Analysis of Budget, Revenue, Genre, and Language

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Introduction

Context

The global film industry generates substantial revenue through cinema releases and streaming platforms, and understanding what drives a film's success have real commercial value for studios, streaming platforms, and investors alike. This report analyses a dataset of movies released between 2000 and 2024, sourced from the Movie Database(TMDB),to explore the factors associated with commercial and critical success. The analysis follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, structured across two iterative cycles (Bemthuis et al., 2025). The first cycle examines broad, global trends across the full dataset - release volume, ratings, revenue, and budget over time. The second cycle builds on these findings with a more focused investigation into how genre and language relate to film's commercial and critical performance.

Research Objectives

First Cycle

- To identify overall patterns in movie release volume, ratings, and revenue between 2000 and 2024.
- To examine the relationship between a movie's budget and the revenue it generates
- To identify which movies achieved the highest revenue and the highest ratings, and to understand which features are most associated with movie success

Second Cycle

- To examine how a movie's genre relates to its ratings and commercial success
- To examine how a movie's original language relates to its ratings and commercial success

First CRISP-DM Cycle

Business Understanding

Rising production costs and tighter budgets have made financial sustainability an ongoing challenge for the film industry. Understanding how budget relates to a film's eventual success is therefore a key business question. This cycle focuses on that business context from the perspective of a specific stakeholder, breaking the overall question down into the sub-questions addressed below.

Stakeholder

This analysis is framed for a **streaming platform content-acquisition team**, responsible for deciding which types of films to license, promote, or commission. This stakeholder needs to understand which factors reliably predict commercial success (revenue) and critical/audience reception (rating), so that acquisition and production-partnership decisions can be prioritized accordingly.

Success Criteria

This analysis would be deemed a success if the following three factors were achieved: (1) identification of consistent patterns in the relationship between budget, genre, release timing, and both rating and revenue; (2) determining whether high ratings consistently correlate with high revenue or whether the two are relatively independent of each other; and (3) practical advice based on the data analysis for the stakeholder.

Research Questions

- What are the key patterns of the success factors of movies in the dataset used?
- How is the movie budget connected to the movie revenue in the film industry?
- What type of changes have the movies experienced over time in terms of release, revenue and rating patterns?
- Which movies have received the highest ratings and have generated the highest revenue?

Data Understanding

The dataset comprises 25 years of CSV files, one for each year from 2000 to 2024, having common columns like id, title, genres, release date, budget, revenue, vote average, vote count, and original language. These 25 years of datasets have been merged together in one single file for analysis. The reason for selecting these specific features is that they reflect answers to the stakeholder's main concerns: budget, revenue, and profit relate to the commercial success, while vote average and vote count refer to critical and audience reception, respectively. Release date, genres, and original language help in analyzing success with the passage of time and according to the content type.

Number of rows: 670208

Number of columns: 25

Table 1: Sample of Raw Merged Data

id	title	vote_average	vote_count	release_date	revenue	budget	runtime	original_language
98	Gladiator	8.209	16962	2000-05-04	465361176	1.03e+08	155	en
77	Memento	8.188	13723	2000-10-11	39723096	9.00e+06	113	en
8358	Cast Away	7.662	10531	2000-12-22	429632142	9.00e+07	143	en
36657	X-Men	6.996	10530	2000-07-13	296339527	7.50e+07	104	en
1359	American Psycho	7.400	9797	2000-04-13	34266564	7.00e+06	102	en

id	title	vote_average	vote_count	release_date	revenue	budget	runtime	original_language
641	Requiem for a Dream	8.027	9165	2000-10-06	7390108	4.50e+06	102	en

The new dataset comprise 670,208 rows and 25 columns. The high number of observations over a 25-year period makes it possible to look for consistent patterns rather than just exceptions.

Data Preparation

Columns were standardized and the columns were transformed into their respective data types (e.g., date-time values, numeric values, Boolean values). Duplicates in the dataset based on the movie id were dropped, resulting in 670,072 observations after cleaning. Other variables that were computed during this step were profit, Return On Investment (ROI), revenue category, and rating category, with filtering done for movies made from 2000-2024. Missing value removal was not performed in this stage because removing them will make the dataset much smaller; instead, the missing values for budget/runtime are corrected to NA because invalid values (negative or zero budget/runtime) can distort our calculations.

Table 2: Missing Values by Column

Variable	Missing_Count
roi	627849
homepage	572950
tagline	567864
keywords	468421
backdrop_path	450480
production_companies	344332
imdb_id	343030
production_countries	273520
spoken_languages	252551
genres	239355
runtime	147612
poster_path	143665
overview	110744
title	4
original_title	4
revenue	1
profit	1
source_file_id	0
id	0
vote_average	0
vote_count	0
status	0
release_date	0
adult	0
budget	0
original_language	0
popularity	0
release_year	0
release_month	0
release_day	0
revenue_success	0

Variable	Missing_Count
rating_group	0

Modeling

Global Trends

After merging and cleaning the dataset, a simple statistical description offers a preliminary overview of the data to be analyzed further. The mean rating for 670072 movies produced in the period from 2000 to 2024 is 2.050817, while the mean revenue is 894739.4; the average budget equals 351558.2 and the mean duration is 68.77461.

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##
## ===== BASIC SUMMARY =====

## Total movies: 670072

## Year range: 2000 to 2024

## Average rating: 2.050817

## Average revenue: 894739.4

## Average budget: 351558.2

## Average runtime: 68.77461
```

To identify broad patterns before narrowing the focus in Cycle 2, the top-performing movies by revenue and rating were tabulated below, alongside yearly release counts and average revenue, with the corresponding trends examined in the plots that follow.

Table 3: Top 10 Revenue Generating Movies

title	release_year	revenue	budget	profit	vote_average	popularity
Adventures in Bora Bora	2023	3000000000	8.00e+08	2200000000	0.000	0.000
TikTok Rizz Party	2024	3000000000	2.50e+08	2750000000	10.000	0.000
Avatar	2009	2923706026	2.37e+08	2686706026	7.573	79.932
Avengers: Endgame	2019	2800000000	3.56e+08	2444000000	8.263	91.756
Avatar: The Way of Water	2022	2320250281	4.60e+08	1860250281	7.654	241.285
Star Wars: The Force Awakens	2015	2068223624	2.45e+08	1823223624	7.293	66.772
Avengers: Infinity War	2018	2052415039	3.00e+08	1752415039	8.255	154.340
Best Of Joy	2024	2000000000	1.00e+03	1999999000	0.000	0.000
Spider-Man: No Way Home	2021	1921847111	2.00e+08	1721847111	7.990	186.065
Jurassic World	2015	1671537444	1.50e+08	1521537444	6.682	54.089

Table 4: Top 10 Highest Rated Movies

title	release_year	vote_average	vote_count	revenue	popularity
Prison High Pressure	2019	9.200	210	0	297.563
BTS World Tour: Love Yourself - Japan Edition	2019	9.172	306	0	4.892
Break the Silence: The Movie	2020	9.154	175	8954945	7.198
BTS Permission to Dance On Stage - Seoul: Live Viewing	2022	9.018	113	32600000	6.877
Franco Escamilla: por la anécdota	2018	9.000	118	0	7.013
BTS: Permission to Dance on Stage - LA	2022	8.993	148	0	11.480
The Three Deaths of Marisela Escobedo	2020	8.904	215	0	8.083
Metallica & San Francisco Symphony: S&M2	2019	8.759	110	0	6.698
Chivas: The Movie	2018	8.655	113	0	5.148
BLACKPINK: Arena Tour 2018 'Special Final in Kyocera Dome Osaka'	2019	8.652	141	0	4.811

Table 5: Count of Movies Per Year

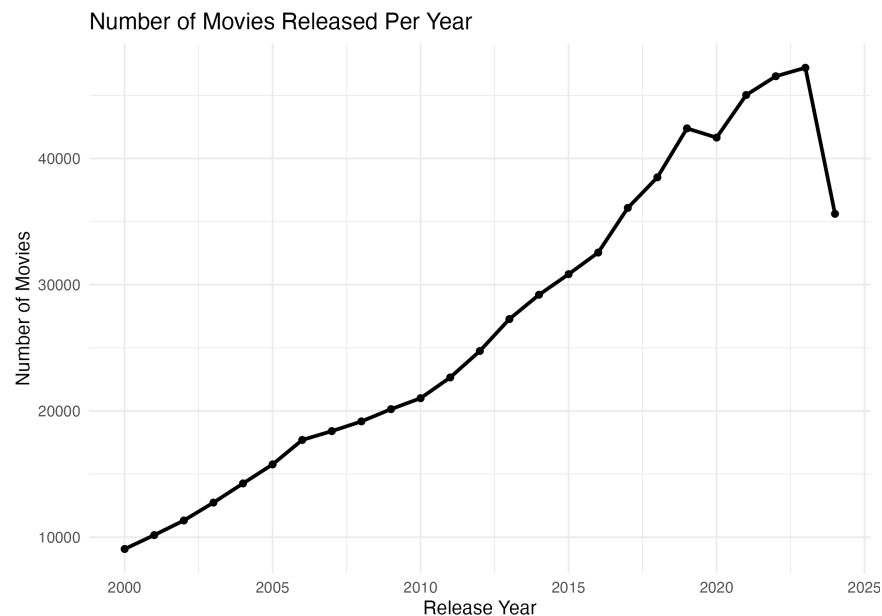
release_year	n
2000	9066
2001	10171
2002	11326
2003	12743
2004	14262
2005	15771
2006	17704
2007	18407
2008	19175
2009	20145
2010	21016
2011	22658
2012	24751
2013	27283
2014	29207
2015	30837
2016	32551
2017	36088
2018	38511
2019	42386
2020	41658
2021	45030
2022	46518
2023	47190
2024	35618

Table 6: Average Revenue Per Year

release_year	avg_revenue	total_movies
2000	1421973.1	9066
2001	1537335.3	10171

release_year	avg_revenue	total_movies
2002	1575080.9	11326
2003	1524395.7	12743
2004	1350951.7	14262
2005	1179745.8	15771
2006	1174869.8	17704
2007	1233752.5	18407
2008	1256332.4	19175
2009	1306485.2	20145
2010	1209525.3	21016
2011	1273173.2	22658
2012	1153215.4	24751
2013	1096160.9	27283
2014	1041021.4	29207
2015	1020166.7	30837
2016	1043425.5	32551
2017	943972.4	36088
2018	856827.6	38511
2019	759689.7	42386
2020	215126.1	41658
2021	378676.5	45030
2022	549097.4	46518
2023	574136.8	47190
2024	435081.4	35618

The number of movies produced every year increased gradually from 9,066 in the year 2000 to reach its highest point of 47,190 in the year 2023, followed by a decrease to 35,618 movies in 2024. A slight decrease can be observed in the year 2020 (41,658 movies compared to 46,518 in 2019). This might be due to a decline in the production and release of movies in the year 2020 due to the pandemic outbreak of COVID-19, as it recovered by the year 2021-2023. The steep fall after 2023 is much more than the fall observed in 2020.

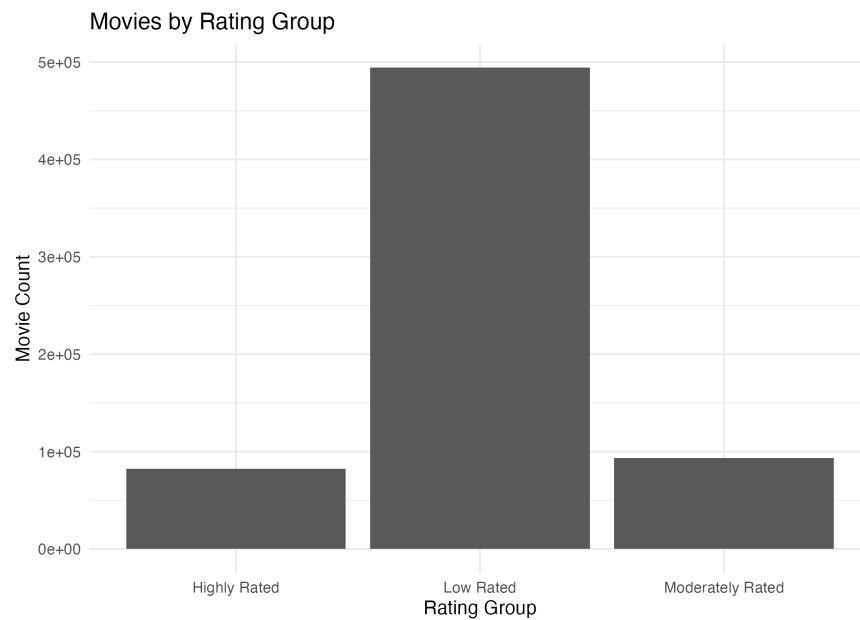
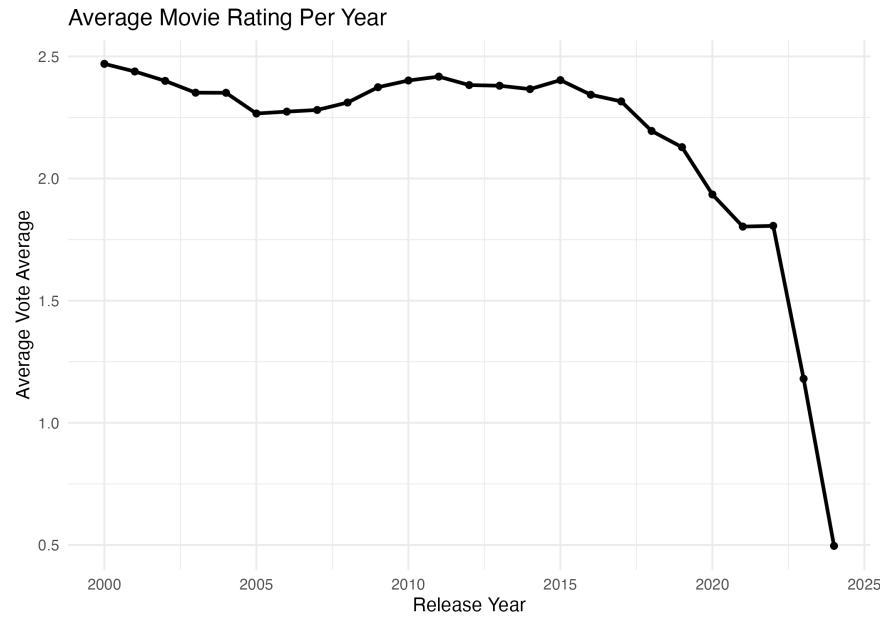


Average movie rating has dropped from 2.47 in the year 2000 to 0.50 in the year 2024, with the largest drop recorded after the year 2020 (from 2.15 in 2020 to 1.18 in 2023, and further down to 0.50 in 2024).

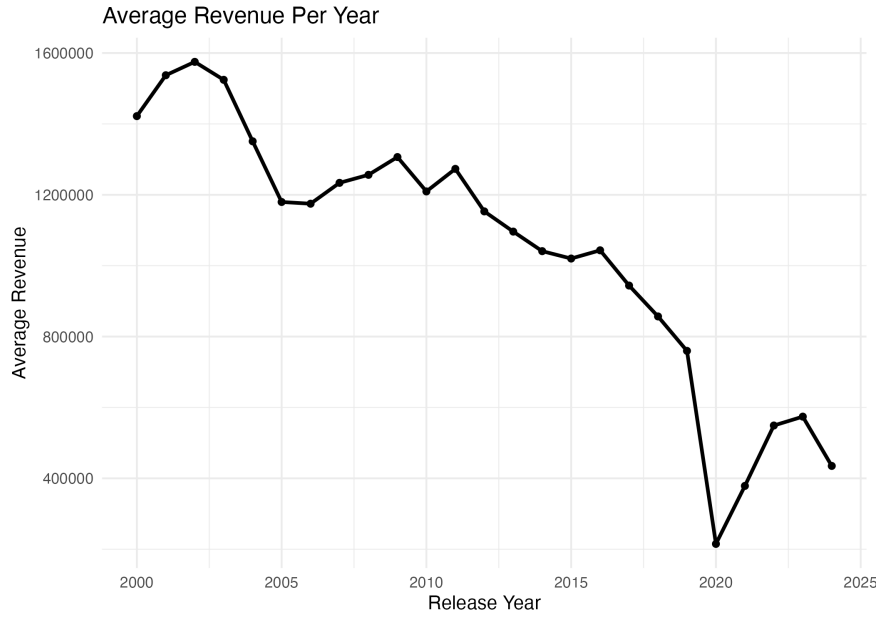
Nevertheless, it would be better to interpret the above finding cautiously and not take it as direct evidence of declining movie quality. According to the Movies by Rating Group graph presented below, most of the movies in the dataset belong to the Low Rated group, and the TMDb’s bulk data contains numerous low engagement or incomplete entries – some of which definitely indicate data quality problems rather than the release of new movies (see Study Limitations). Thus, the above mentioned decline in average ratings may indicate a growing share of these low-quality data among the more recent data.

Table 7: Average Ratings Per Year

release_year	avg_rating	n_movies
2000	2.4696467	9066
2001	2.4384116	10171
2002	2.3999282	11326
2003	2.3516138	12743
2004	2.3511642	14262
2005	2.2663757	15771
2006	2.2740765	17704
2007	2.2808919	18407
2008	2.3112832	19175
2009	2.3740385	20145
2010	2.4015279	21016
2011	2.4173915	22658
2012	2.3827337	24751
2013	2.3801407	27283
2014	2.3662825	29207
2015	2.4030771	30837
2016	2.3432501	32551
2017	2.3160067	36088
2018	2.1949464	38511
2019	2.1285562	42386
2020	1.9348973	41658
2021	1.8033391	45030
2022	1.8064311	46518
2023	1.1804539	47190
2024	0.4960628	35618

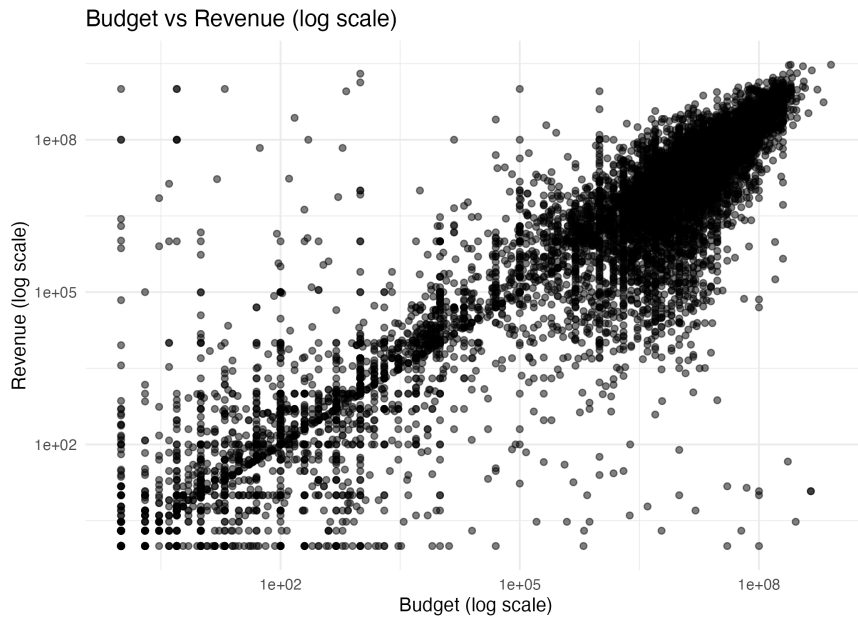


In line with the above trend, low-rated films greatly outweigh films that are of medium and high ratings, collectively making up about three-quarters of the entire dataset. This provides additional evidence that the decrease in the average rating with time is an indication of an increasing proportion of low-engagement and low-quality films.

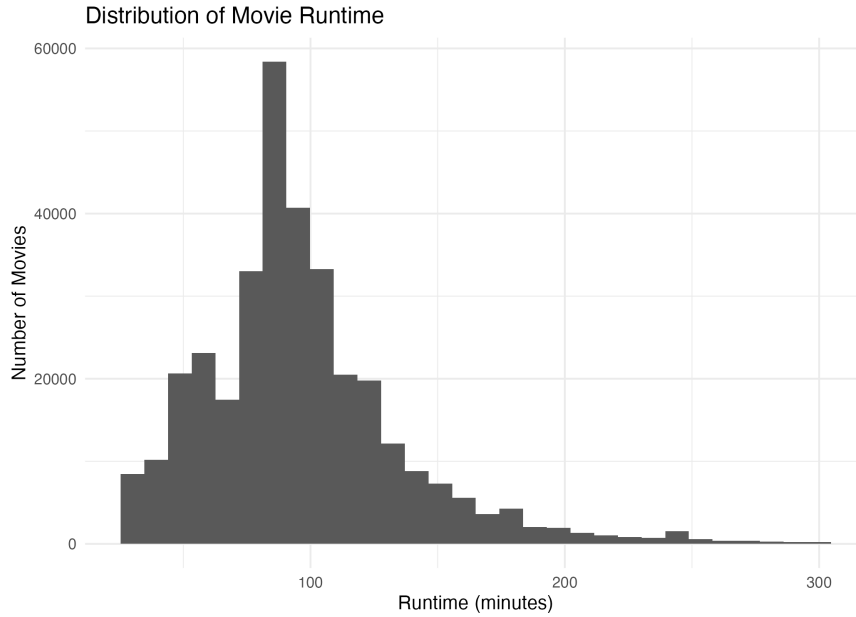


The average annual revenues went up from 1,421,973 in 2000 to 1,575,081 in 2002 and then fell steadily to 759,690 in 2019. There was a sudden fall in 2020 to 215,126, which can be attributed to the pandemic-driven closure of movie theatres, after which there was a slight bounce back in 2023 to 574,137 and then another fall to 435,081 in 2024.

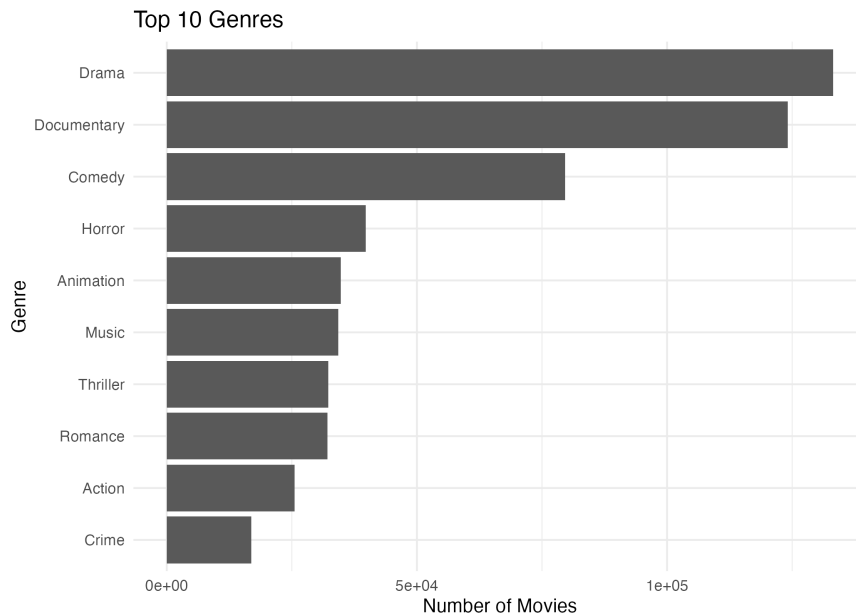
On the whole, the long-run trend throughout the entire period depicts falling average revenue, in line with the trend witnessed in the ratings presented above.



Drama, Documentary, and Comedy are the three most common genres by movie count, together accounting for a substantial share of all releases in the dataset. This indicates that recent film production continues to concentrate heavily on these established genres. Whether this high production volume translates into stronger commercial or critical performance is examined further in Cycle 2.



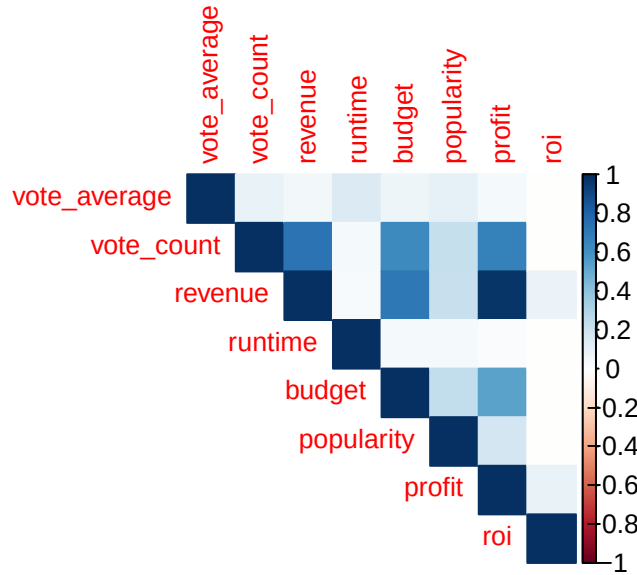
Most films have their duration distributed around a region between approximately 80 and 120 minutes, reflecting feature films' durations. Some films are much longer than this interval and constitute the tail of longer durations.



The correlation between revenue and profit is extremely high (0.97), which is understandable because profit is calculated from revenue. The correlation between the number of votes and revenue is also quite high (0.73), as well as the number of votes and the budget of the movie (0.64). However, this refers to the number of people who saw or voted for the movie, and not to how good the movie was because the correlation between the average value of the rating (`vote_average`), on the one hand, and revenue (0.06), budget (0.08) and profit (0.05), on the other hand, is extremely low. This means that the success and popularity of the movie are largely independent of its reception: a high rating does not necessarily lead to a high revenue and vice versa. Popularity is positively correlated with revenue (0.22) and the number of votes (0.23).

Table 8: Correlation Matrix

	vote_average	vote_count	revenue	runtime	budget	popularity	profit	roi
vote_average	1.0000000	0.0958849	0.0596094	0.1538229	0.0756965	0.1006427	0.0469618	-
vote_count	0.0958849	1.0000000	0.7300981	0.0456980	0.6356562	0.2327055	0.6729876	-
revenue	0.0596094	0.7300981	1.0000000	0.0346697	0.7195390	0.2229483	0.9724761	-
runtime	0.1538229	0.0456980	0.0346697	1.0000000	0.0498787	0.0416310	0.0256571	-
budget	0.0756965	0.6356562	0.7195390	0.0498787	1.0000000	0.2459807	0.5379253	-
popularity	0.1006427	0.2327055	0.2229483	0.0416310	0.2459807	1.0000000	0.1881045	-
profit	0.0469618	0.6729876	0.9724761	0.0256571	0.5379253	0.1881045	1.0000000	-
roi	-0.0071292	-	0.0817757	-	-	-	0.0999519	1.0000000
		0.0016258		0.0087336	0.0019635	0.0009876		



Evaluation - Cycle 1

Key Findings

- Number of movies released per year increasing.
- Average rating of movies per year decreasing.
- Average revenue fluctuating per year but tending to decrease steadily.
- Budget has a strong relationship with revenue.
- High ratings do not imply high revenue.

Link to Success Criteria

These findings speak directly to the success criteria set out in the Business Understanding: a clear pattern has been identified linking budget to revenue, while the connection between ratings and revenue is shown to be negligible - meaning ratings and revenue should be treated as largely independent signals of success rather than one predicting the other. This gives the stakeholder an early, data-grounded basis for acquisition decisions, while highlighting that further investigation (carried out in Cycle 2) is needed before more specific, actionable guidance can be given.

Limitations

Negligible connection between different factors and movie success has been identified.

Presence of missing values in different columns that are included in the analysis process.

Questions for Cycle 2

Which movie genre generates highest revenue and ratings?

Does genre and language have any special influence on movie success?

Second CRISP-DM Cycle

Refined Business Understanding

Based on the overall data analysis and findings received from the first CRISP-DM Cycle, it has been understood that most of the mentioned factors are interconnected to each other moderately. In this regard, the study intends to understand specifically, the connection between genre and revenue of movies in detail in the second CRISP-DM Cycle. This cycle also intends the way language influence the revenue of movies.

Research Questions

What is the best movie genre in the dataset based on the revenue and rating factors?

How does the movie language influence the success of movies?

Enhanced Data Preparation

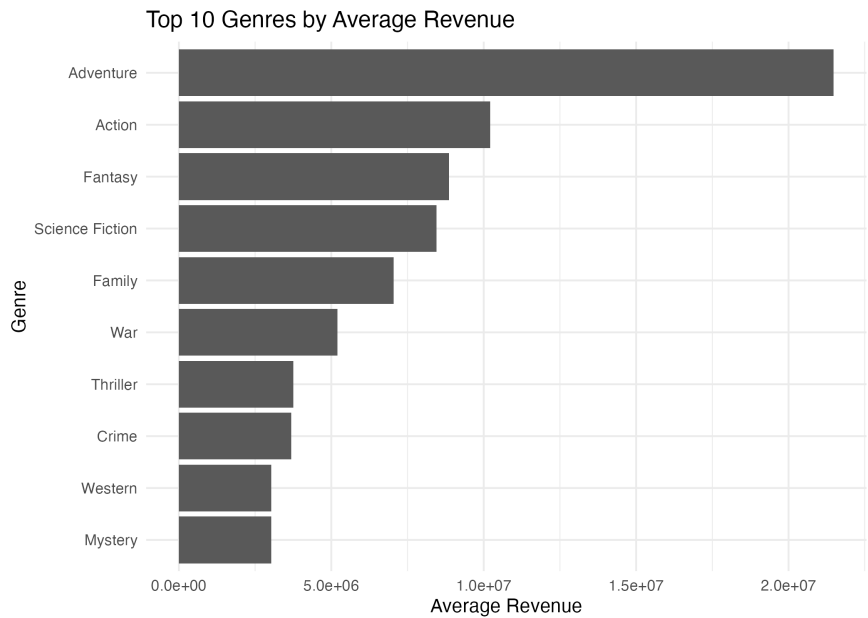
The data has been prepared for this section by selecting only the required columns from the dataset, such as revenue, genres, title, vote average, and id. The rows containing missing values in the genre columns have been removed from the dataset. The separate rows have been created based on separate genres to analyse the impact of individual genres on movie success. The columns have been created called average rating, average revenue and movie count, calculating their values from the dataset. The numerical columns have been sorted in decreasing order to identify the best genre.

Advanced Modeling and Final Evaluation

Influence of Genre on Movie Success

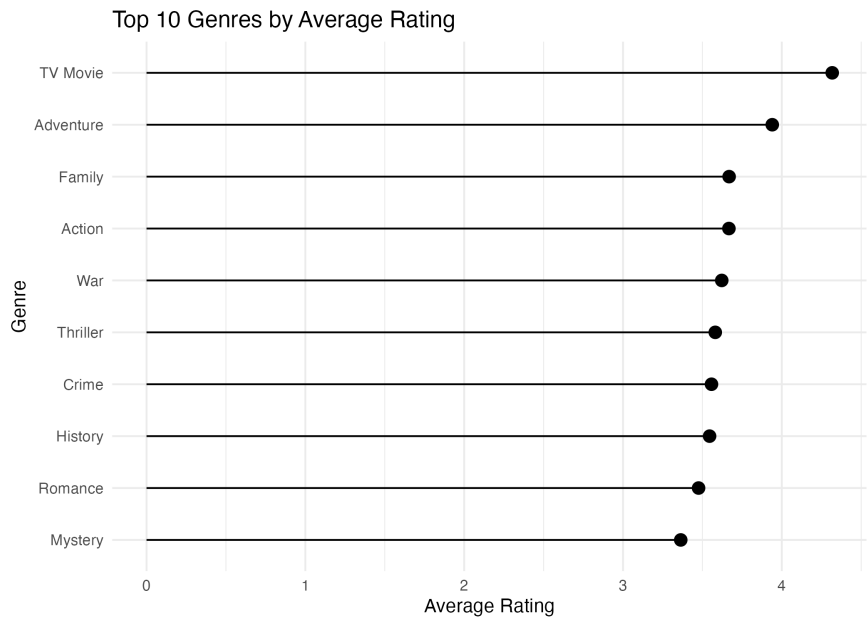
The exploration of the table comparing different genres in terms of revenue, ratings and others reveals that adventure is the best genre among the existing genres as it has contributed to the highest average revenue and average ratings. However, the highest movie count has been noticed in the drama genre, indicating that

the emphasis on drama genre for movie development is one of the significant causes of deteriorating ratings and revenue from 2000 to 2024 in the film industry. Hence, the film producers need to emphasise creating adventure movies to gain high ratings, revenue and movie success.



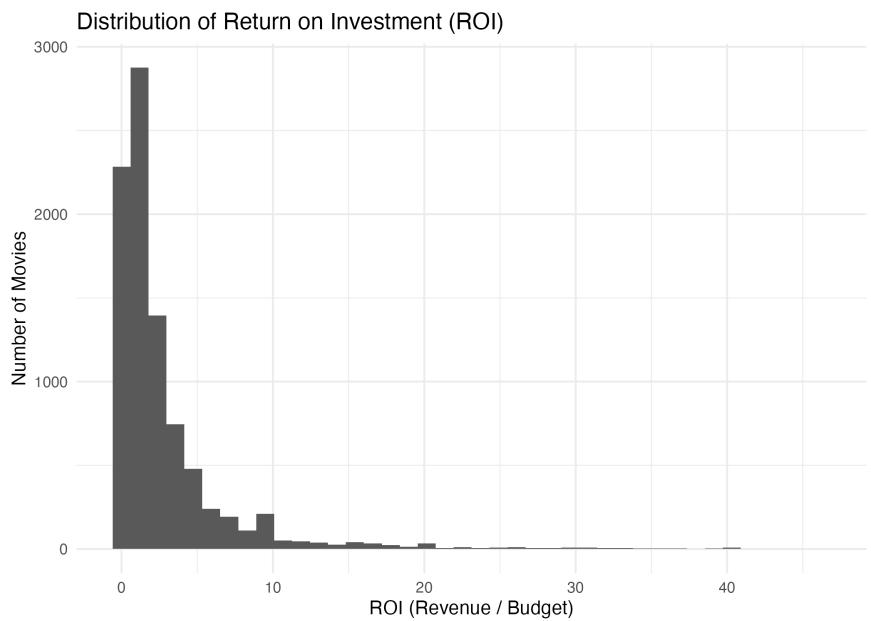
The revenue and ratings may be considered the most important factors to estimate movie success, as high ratings by audiences can gradually attract more audiences to visit the hall to watch a movie. Revenue, being strongly connected to audience count, also depends on ratings, as the more the ratings the more the crowd of audience, indicating that ratings are the prime factor for movie success. Based on revenue, adventure is the genre generating highest revenue per year.

To go beyond revenue alone, the genres have also been ranked by average rating, and the overall spread of Return on Investment (ROI) across all movies has been examined, giving a fuller picture of genre performance and production efficiency.

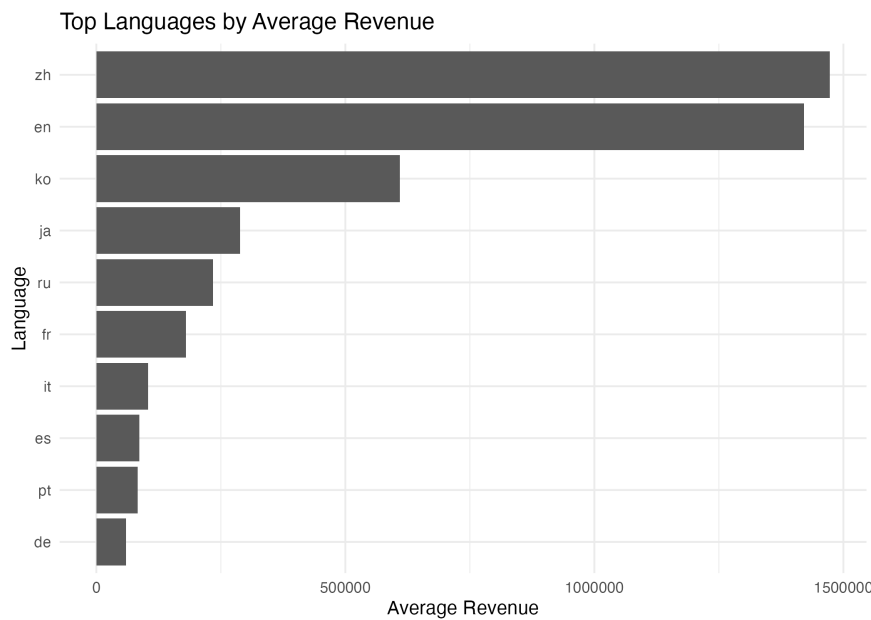


The distribution of Return on Investment shows that most movies cluster at low-to-moderate ROI, with a

long tail of high-ROI outliers, indicating that a small number of movies generate disproportionately large returns relative to their budget.

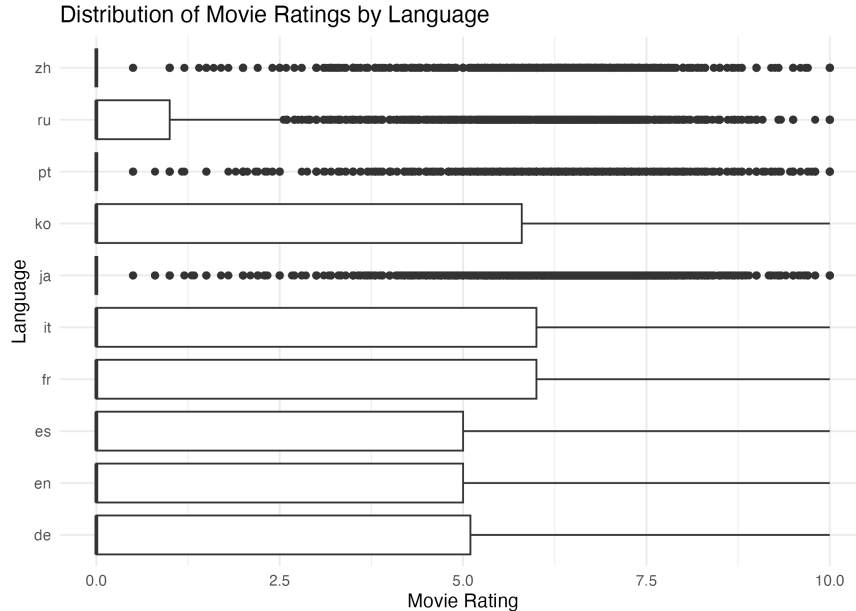


Influence of Language on Movie Success



The analysis shows that the English language movies, despite generating high revenue and counts, possess limited ratings. On the other hand, the movies in other languages such as Chinese and other languages, despite having lower movie counts than English, have received competitive ratings. This indicates that the complete emphasis on the English language only by production companies may not be an appropriate decision to enhance movie success. It should be noted that this comparison is restricted to the 10 most common languages by movie count, so as to avoid unstable averages from languages with very few titles.

Connection between language and ratings



The Italian and French languages has been found to be at the top, indicating that audiences prefer these languages more than other languages. In this regard, developing movies in these languages can help gain huge audience count enhancing the chances of movie success.

Evaluation - Cycle 2

Key Findings

Adventure is the genre most strongly associated with both high average revenue and high average ratings.

Drama has the highest movie count but does not lead on revenue or rating, suggesting oversupply relative to audience reward.

English and Chinese language movies generate the highest revenue, but Italian and French language movies achieve the highest average ratings.

Revenue and ratings are not strongly linked - the highest-revenue movies are rarely the highest-rated ones.

Link to Success Criteria

These findings directly satisfy the success criteria set out in the Business Understanding: a clear pattern has been identified (Adventure genre and non-English languages both associate with stronger audience reception), and the revenue-rating relationship has been clarified as largely independent rather than reliably predictive. This gives the stakeholder a concrete basis for content decisions: prioritising Adventure titles for revenue, while considering multilingual content investment to strengthen ratings.

Limitations

Genre and language are both self-reported, categorical fields in the source data and may not capture nuanced audience segments.

The analysis does not control for confounding factors such as marketing spend or release timing, which likely also influence genre and language performance.

Next Steps

Future work could model revenue and rating jointly (e.g. multivariate regression) to isolate the independent effect of genre and language while controlling for budget and release year.

A dedicated success label (rather than deriving it from revenue/rating thresholds) would allow more precise classification-based analysis in a future CRISP-DM cycle.

Final Evaluation

Major Findings

English and Chinese language movies have mostly generated high revenue.

Adventure genre movies has been found to generate significantly high revenue.

Italian and French language are top-rated languages among others.

Revenue and ratings are not directly connected to each other as the highest revenue movie and top-rated movies are different.

Implications for Film Production Industry

Focus on generating movies on multiple languages can help film producers equally gain high ratings and revenue.

The adventure genre has been found to generate highest revenue indicating the film production industry can focus on this genre to enhance profitability.

Developing movies in Italian and French languages can enhances audience count, enhancing revenue.

The emphasis on multiple languages can help create movies in native languages of different people enhancing their understandability resulting in increased audience.

High ratings does not ensure high revenue, indicating that a high quality movie content is a priority along with top-rated language to ensure a successful movie.

Study Limitations

The dataset consists of huge missing values in some of the columns which has been challenging to handle as the removal of those rows might have reduced the data size, imposing negative impact to the data analysis.

The findings of the study has been extracted from a single dataset.

The dataset lacks a separate column specifically designed to label movies as successful and failure. A small number of entries in the raw data (e.g. movies with several billion in reported revenue against a budget of only a few hundred dollars and zero votes) also appear to be data quality issues rather than genuine box-office performance, and should be interpreted with caution where they influence top-revenue rankings.

Conclusions

Based on the overall analysis in the CRISP-DM Cycles, it can be inferred that the emphasis on creating more movies in the adventure genre is necessary for enhanced profitability and revenue increase in the film industry. The ratings are the most important factor influencing the success of a movie by spreading positive vibes and attracting new audiences to watch the movie. The development of movies with top-rated languages such as Italian, French and others can influence movies success. The film industry can rely on this information to enhance its growth and revenue in future.

Methodological Notes

The CRISP-DM methodological approach has been executed with two iterative cycles.

The first cycle has focused on exploring global trends.

The second cycle has focused on specific details regarding the connection of movie success with genre, language, and ratings.

The R programming language has been used to execute the entire coding part.

Histogram, bar plots and others have been used to analyse the dataset. All exploratory and modelling visualisations in this report were produced using ggplot2, with a single exception: the correlation matrix visualisation uses the corplot package, as it produces a purpose-built, directly interpretable colour-coded correlation grid that ggplot2 does not offer natively without substantial additional code.

References

- Bemthuis, R. H., Govers, R. R., & Asadi, A. (2025). A CRISP-DM-based methodology for assessing agent-based simulation models using process mining [Review of A CRISP-DM-based methodology for assessing agent-based simulation models using process mining]. *Journal of Simulation*, 1-22. <https://doi.org/10.1080/17477778.2025.2508245>
- Singh, P., & Asthana, S. (2025). Financial Sustainability in the Film Industry: Challenges, Strategies, and Future Directions. *International Journal on Science and Technology (IJSAT)* IJSAT25033691, 16(3). <https://www.ijسات.org/papers/2025/3/3691.pdf>